

Computer Networks

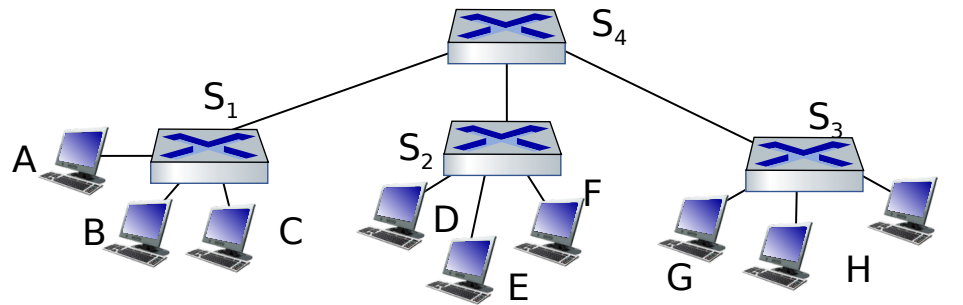
Hubs and Switches

Amitangshu Pal

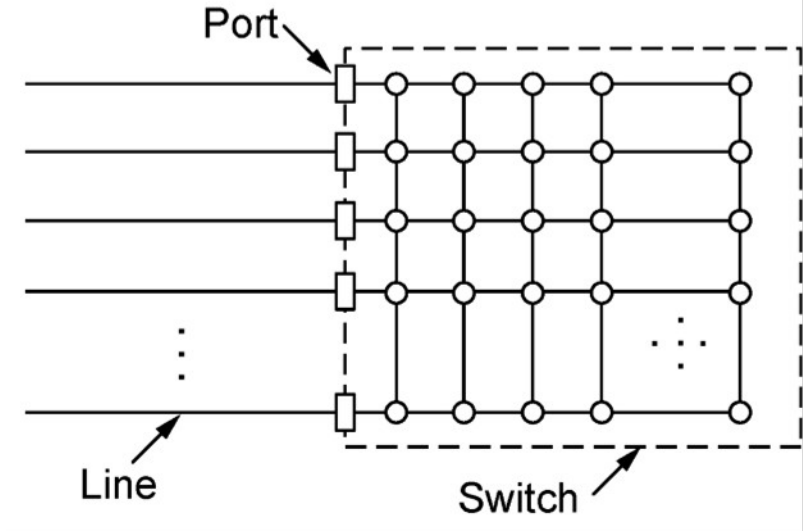
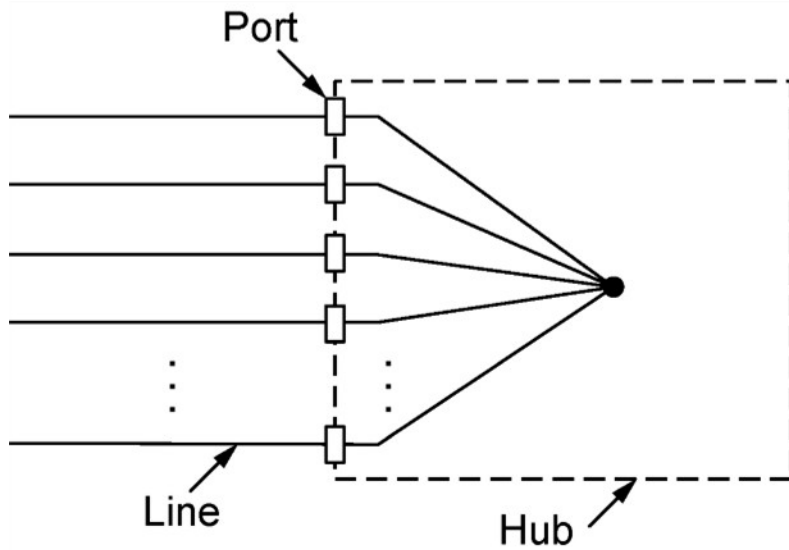
Computer Science and Engineering

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Hubs vs Switches

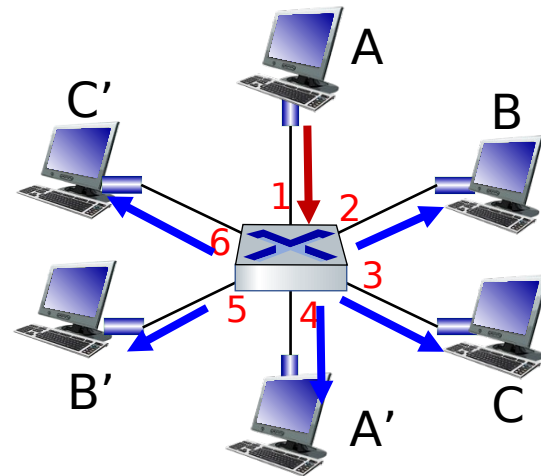


Hubs vs Switches



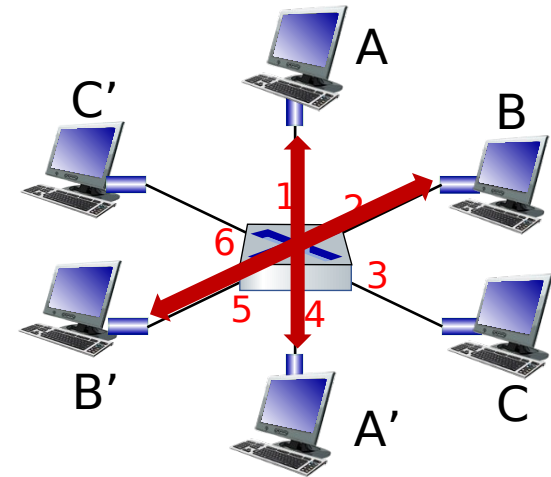
Hubs

- Bits coming from one link is repeated to all other links
- No frame buffering
- No CSMA/CD at hub
 - One large collision domain
- **Hub:** A-to-A' and B-to-B' cannot transmit simultaneously



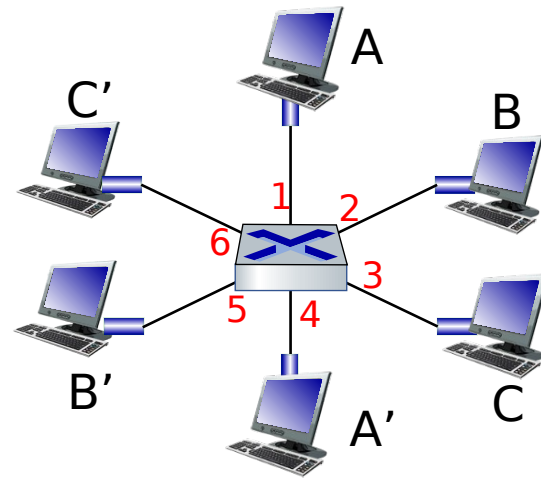
Switches

- Hosts have dedicated, direct connection to switch
- Switches buffer packets
- Ethernet protocol used on each incoming link, so:
 - No collisions; full duplex
 - Each link is its own collision domain
- **Switching:** A-to-A' and B-to-B' can transmit simultaneously, without collisions
- Hubs vs switch: <https://community.fs.com/blog/do-you-know-the-differences-between-hubs-switches-and-routers.html>



Cut-through Switches

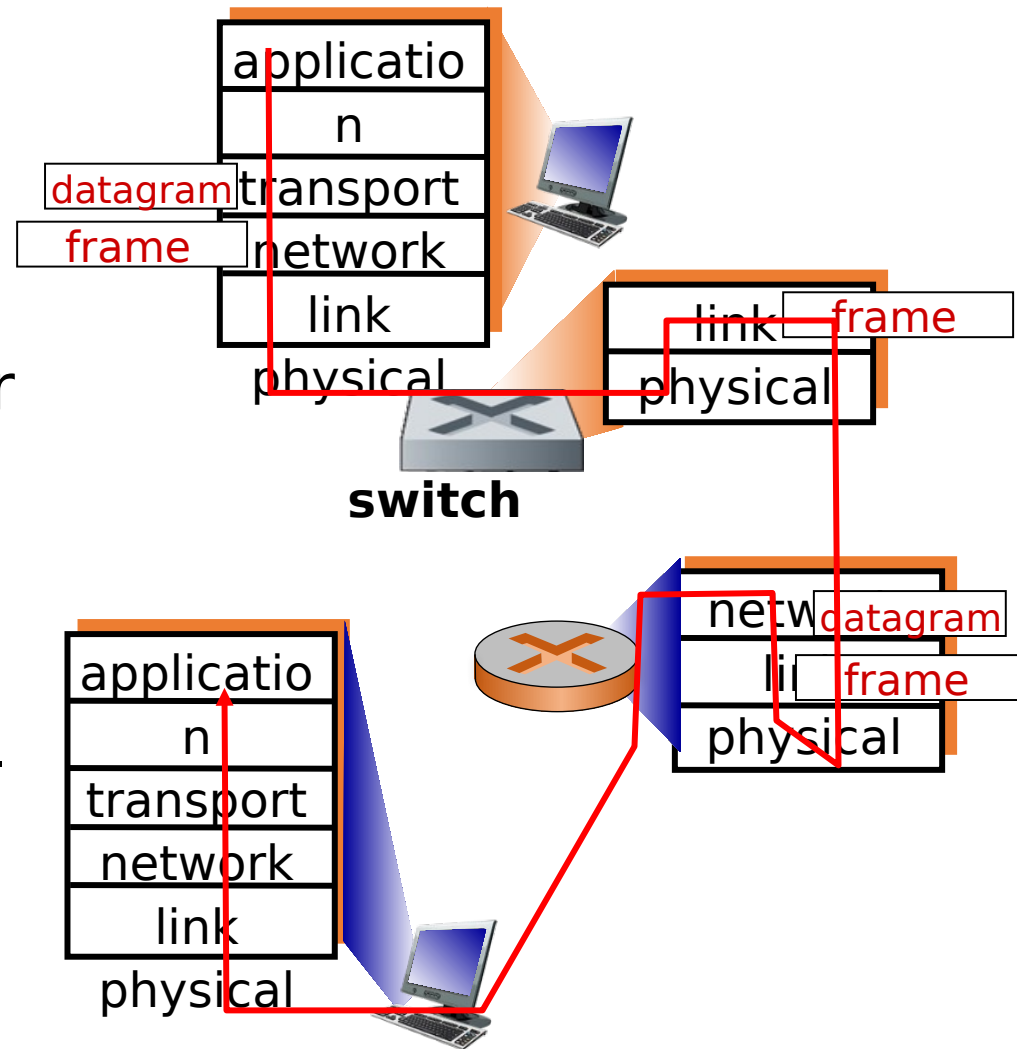
- Switches start forwarding the frames just after reading the destination address
 - Slightly reduces the latency



Switches vs. routers

Both are store-&-forward:

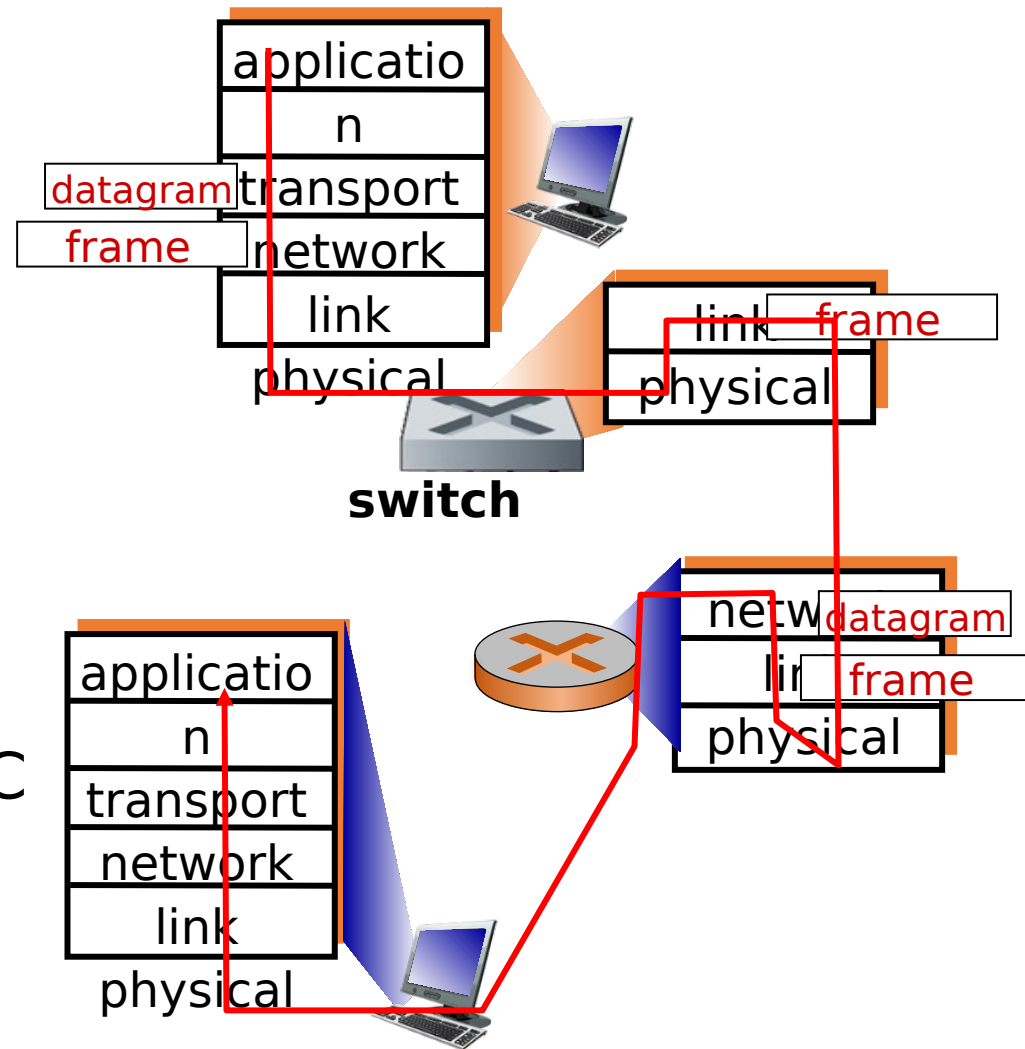
- **Routers:** network-layer devices (examine network-layer headers)
- **Switches:** link-layer devices (examine link-layer headers)



Switches vs. routers

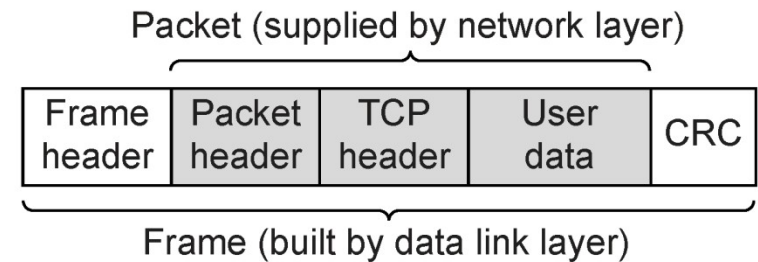
Both have forwarding tables:

- **Routers:** compute tables using routing algorithms, IP addresses
- **Switches:** learn forwarding table using flooding, learning, MAC addresses



Hubs vs Switches vs Routers

Network layer	Router
Data link layer	Bridge, switch
Physical layer	Repeater, hub



Summary

□ Hubs and switches:

- Hubs
 - Switches
 - Cut-through switches
 - Routers
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